EXPERIMENT 6

Aim:
To study Agile Methodology and Test Case Management using JIRA tool and to create a Scrum Board for a Scrum Master using JIRA.

Theory:

Introduction to JIRA:
JIRA is a powerful issue tracking and project management tool developed by Atlassian. It is widely used by software development teams to plan, track, and manage agile software projects efficiently. JIRA helps in organizing tasks, assigning responsibilities, tracking progress, and ensuring timely delivery of projects.

Agile Methodology:
Agile is a software development approach that focuses on iterative development, collaboration, flexibility, and customer satisfaction. Instead of delivering the entire product at once, Agile divides the project into smaller increments called sprints.

Key principles of Agile include:

- Customer collaboration over contract negotiation
- Responding to change over following a plan
- Working software over comprehensive documentation
- Individuals and interactions over processes and tools

JIRA supports Agile frameworks like Scrum and Kanban, enabling teams to manage work in a structured yet flexible manner.

Scrum Framework in JIRA:
Scrum is an Agile framework that divides work into time-boxed iterations called sprints (usually 2–4 weeks). JIRA provides Scrum boards that help teams visualize tasks and track progress.

Important Scrum components:

- **Product Backlog:** List of all tasks/features
- **Sprint Backlog:** Tasks selected for a sprint
- **Daily Standup:** Daily progress meeting
- **Sprint Review:** Demonstration of completed work
- **Sprint Retrospective:** Discussion on improvements

Roles in Scrum:

Scrum Master: Facilitates the process
Product Owner: Defines requirements
Development Team: Executes tasks

Issue Tracking in JIRA:
JIRA uses “issues” to represent tasks, bugs, features, or improvements. Each issue contains details like description, priority, status, and assignee. Issues move through different stages such as:
To Do → In Progress → Done

Workflow Management:
A workflow in JIRA defines the lifecycle of an issue. It consists of statuses and transitions. Teams can customize workflows according to their project needs, making JIRA highly flexible.

Test Case Management in JIRA:
JIRA supports test case management through plugins (like Zephyr or Xray). It helps in:

- Creating and managing test cases
- Linking test cases with requirements
- Executing test cases
- Tracking defects and bugs

This ensures better software quality and traceability.

Scrum Board in JIRA:
A Scrum board is a visual tool that helps teams manage tasks during a sprint. It consists of columns like:

- To Do
- In Progress
- Testing
- Done

Benefits:

- Improves visibility of work
- Tracks progress easily
- Helps identify bottlenecks

Collaboration and Communication:
JIRA enhances teamwork by allowing:

- Comments on issues

· File attachments
· Mentions (@team members)

It also integrates with tools like Confluence, Bitbucket, and Slack for better communication.

Reporting and Analytics:
JIRA provides various reports such as:

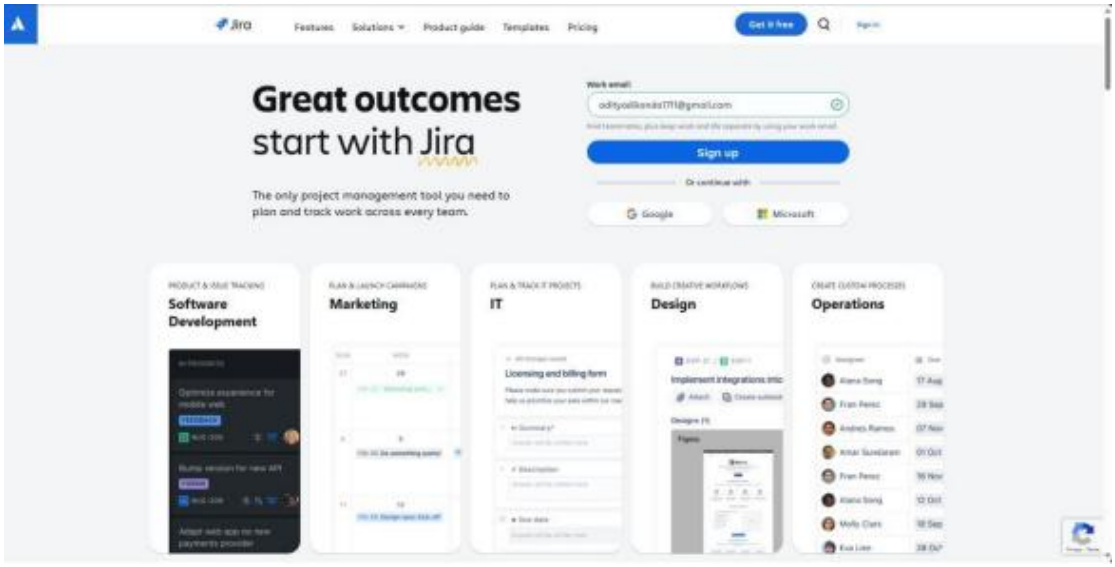
· Burndown Chart
· Velocity Chart
· Sprint Report
· Cumulative Flow Diagram

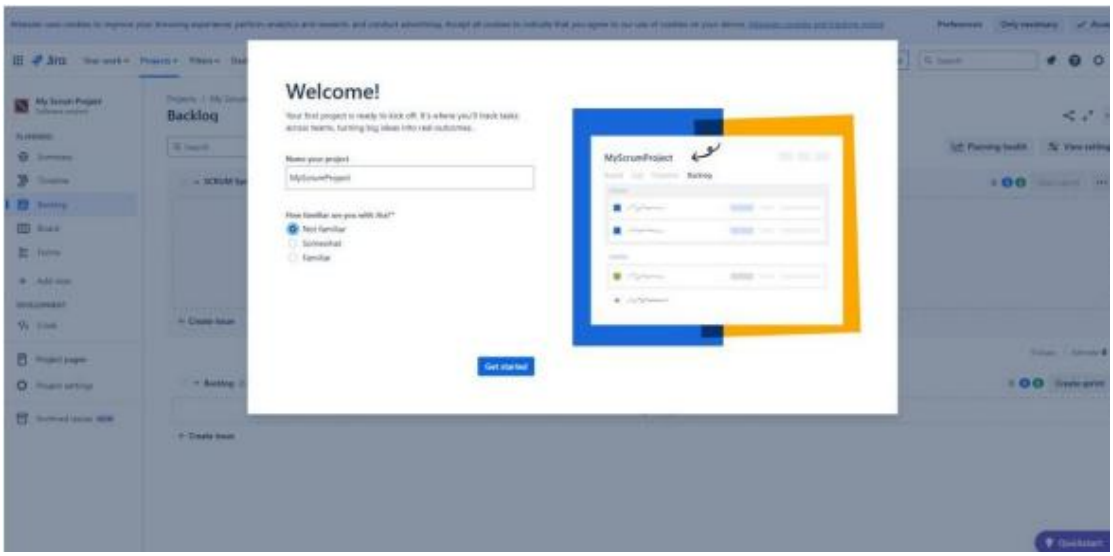
These reports help in analyzing team performance and project progress.

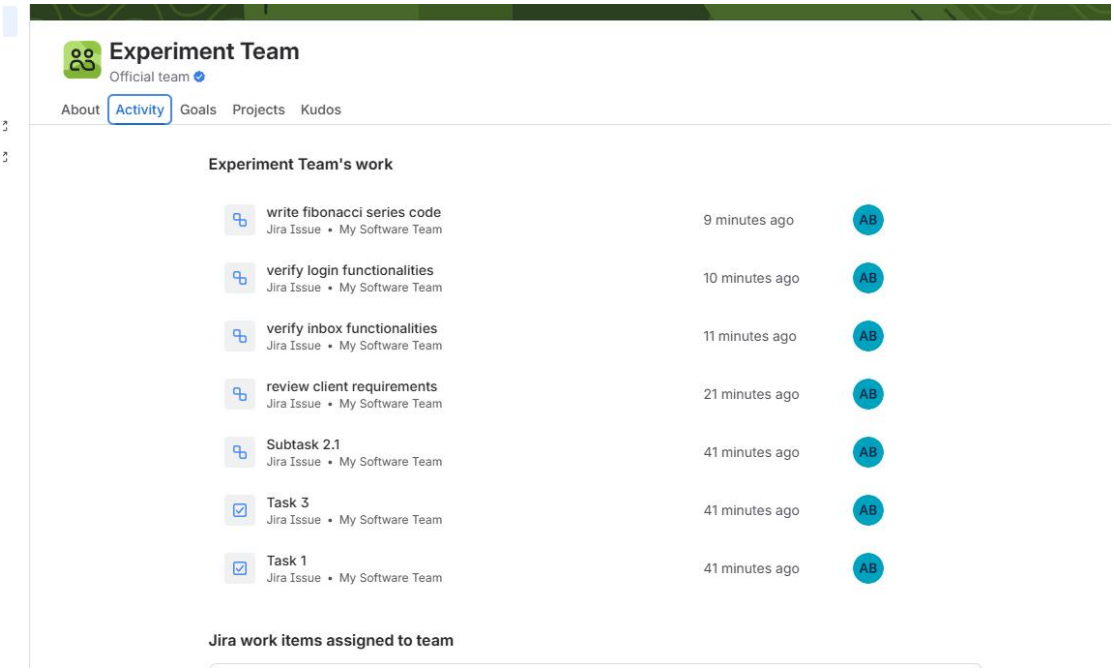
Integration and Scalability:
JIRA integrates with many third-party tools like GitHub, Jenkins, and CI/CD pipelines. It is highly scalable and can be used by small teams as well as large organizations.

Advantages of Using JIRA:

· Easy tracking of tasks and bugs
· Supports Agile methodologies
· Highly customizable
· Improves team collaboration
· Provides detailed reports




Welcome!
Backlog
MyScrumProject
Get started


Experiment Team
Official team
About
Activity
Goals
Projects
Kudos
Experiment Team's work
write fibonacci series code
Jira Issue • My Software Team
9 minutes ago
AB
verify login functionalities
Jira Issue • My Software Team
10 minutes ago
AB
verify inbox functionalities
Jira Issue • My Software Team
11 minutes ago
AB
review client requirements
Jira Issue • My Software Team
21 minutes ago
AB
Subtask 2.1
Jira Issue • My Software Team
41 minutes ago
AB
Task 3
Jira Issue • My Software Team
41 minutes ago
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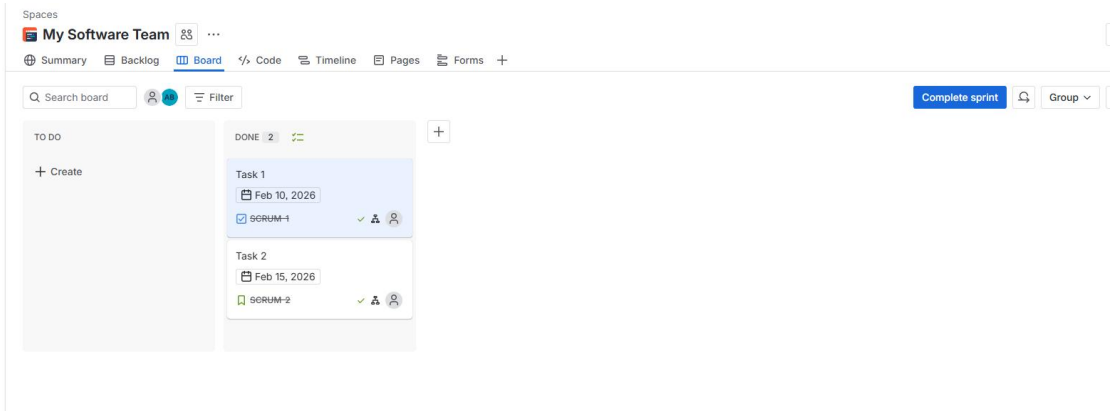
Task 1
Jira Issue • My Software Team
41 minutes ago
AB
Jira work items assigned to team


Spaces
My Software Team
Summary
Backlog
Board
Code
Timeline
Pages
Forms
Search board
Filter
Complete sprint
Group
TO DO
Create
DONE 2
Task 1
Feb 10, 2026
SCRUM-1
Task 2
Feb 15, 2026
SCRUM-2

SCRUM-1 / SCRUM-9

write fibonacci series code

Description

Add a description...

Attachments 1

fibonacci seriescode.c
05 Feb 2026, 02:11 PM

Linked work items

Add linked work item

Activity

All | Comments | History | Work log

AB

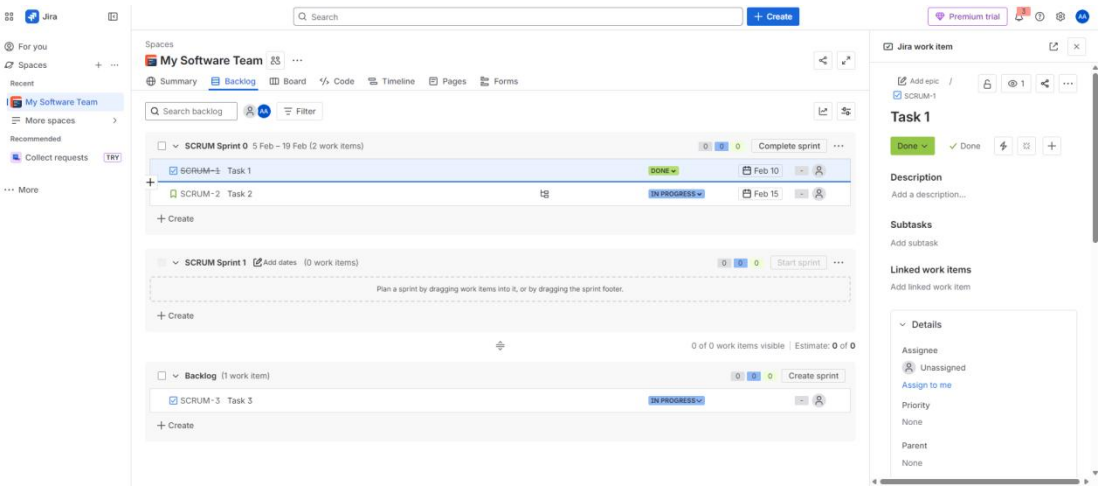
Add a comment...

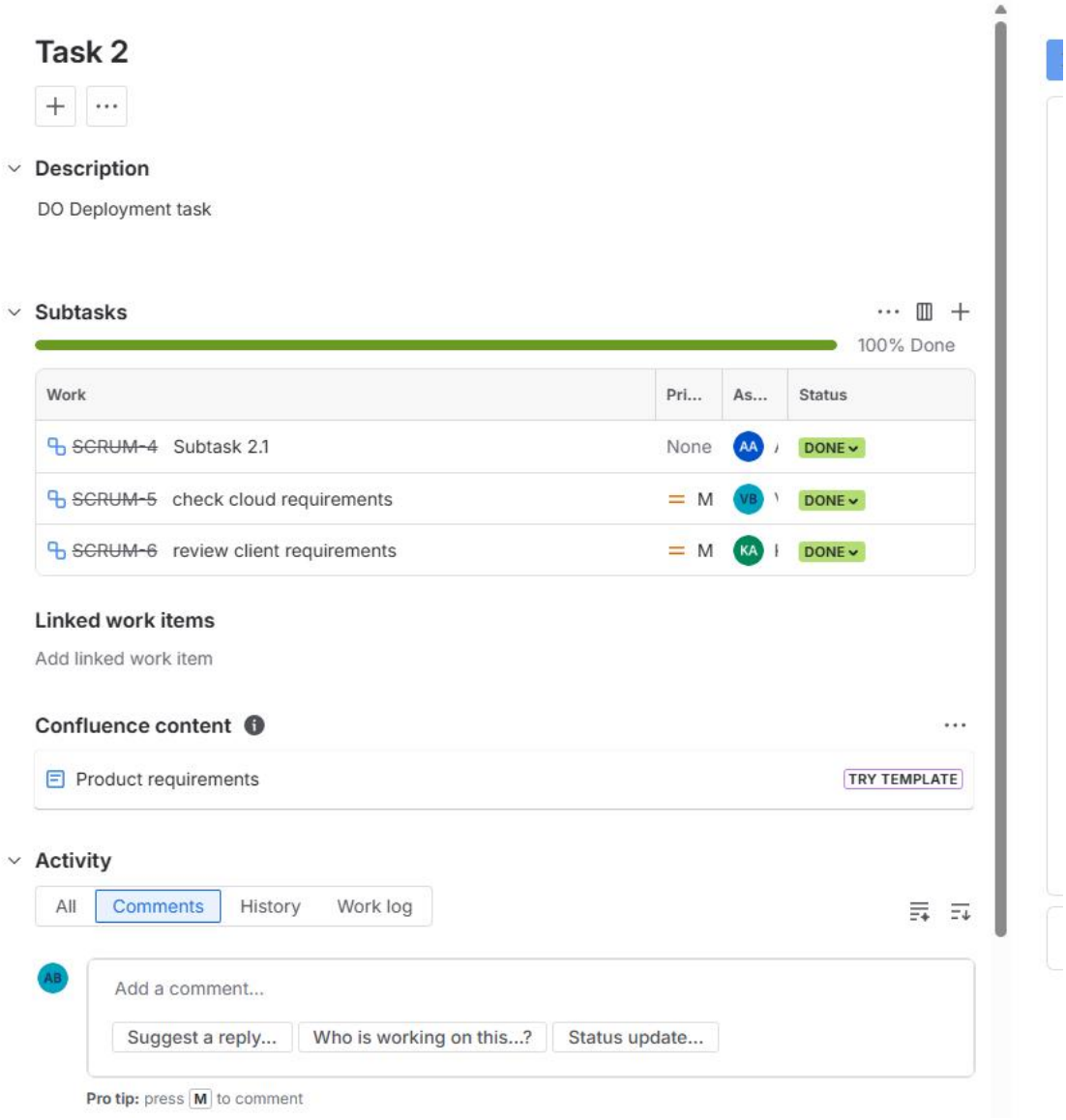
Suggest a reply... | Can I get more info...? | Status update...

Pro tip: press M to comment

AB **Ankita Bedse**
in 1 minute

redo @Vivaan Balchandani





Conclusion:

Thus, we have successfully studied Agile methodology and test case management using the JIRA tool and created a Scrum board for a Scrum Master using JIRA. The tool helps in efficient project management, collaboration, and tracking, making it an essential tool in modern software development.